

Backtest Report for GBT+JEPA Direction, Fixed-Delta 0DTE Execution, and Option Value JEPA Research Models

Kripta Studios

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Abstract

This report documents the current evidence for the GBT+JEPA trading architecture. The strict out-of-sample benchmark trains through March 31, 2026 and tests April 1 through May 29, 2026. In that benchmark, the combined base_jepa 180 minute direction model outperforms both the base GBT and JEPA-only models. The fixed $\Delta = 0.70$ 0DTE option execution rule is currently stronger than the learned OptionValueJEPA strike/exit alternatives on the practical promotion criteria. Production-history diagnostics are reported separately and are explicitly not out-of-sample evidence.

1 Executive Conclusion

The promoted production candidate is:

GBT+JEPA 180 minute direction model + fixed $\Delta \approx 0.70$ 0DTE option execution.

The main reasons are:

1. base_jepa improves strict OOS direction-trade performance relative to the base GBT.
2. JEPA-only is weaker than base+JEPA, so JEPA should be used as a feature generator, not as a standalone replacement.
3. Fixed $\Delta = 0.70$ option execution beats learned strike selectors on the most stable OOS option-policy comparison.
4. Learned 5 minute exits do not yet capture the large oracle-exit ceiling.
5. Production final diagnostics are strong but must be treated as model-history checks, not validation.

2 Evaluation Definitions

2.1 Prediction Metrics

For the 180 minute direction model:

$$y_t = \mathbb{1}[S_{t+180m} > S_t].$$

Reported prediction metrics include:

- AUC,
- accuracy,
- balanced accuracy,
- Spearman correlation between predicted probability and future return,
- top-quintile and bottom-quintile future returns.

2.2 Trade Metrics

For proxy backtests:

$$\text{gross bps}_t = d_t \cdot 10,000 \left(\frac{S_{t+180m}}{S_t} - 1 \right),$$

where $d_t = +1$ for long and $d_t = -1$ for short. Net bps subtracts cost:

$$\text{net bps}_t = \text{gross bps}_t - \text{cost bps}.$$

PnL is:

$$\text{PnL}_t = \frac{\text{net bps}_t}{10,000} \times \text{notional}.$$

Profit factor is:

$$\text{PF} = \frac{\sum_{\text{PnL} > 0} \text{PnL}}{|\sum_{\text{PnL} < 0} \text{PnL}|}.$$

3 Strict OOS 180 Minute Direction Test

3.1 Protocol

- Training cutoff: March 31, 2026.
- Test period: April 1, 2026 through May 29, 2026.
- Tickers: SPX, QQQ, SPY.
- Label: exact 180 minute future direction.
- Hold: fixed 180 minutes.
- Cooldown: 180 minutes.
- Cost: 1 bps.
- Notional: \$100,000 per proxy trade.

This is the key validation split because all training and threshold selection are frozen before the OOS period.

3.2 Prediction Quality

base_jepa has the best AUC and Spearman rank correlation. This matters because trade profitability depends more on ranking and thresholding than on raw accuracy.

Model	Rows	Future Up	Pred Up	AUC	Acc	Bal Acc	Spearman	Top Q bps
base	2,220	62.6%	60.1%	0.626	62.3%	60.4%	0.183	10.82
jepa_only	2,220	62.6%	51.0%	0.579	55.9%	56.0%	0.129	12.29
base_jepa	2,220	62.6%	55.4%	0.636	61.8%	61.1%	0.222	13.89

Table 1: Strict OOS prediction metrics for April–May 2026.

Model	Trades	WR	PF	Avg bps	PnL	Max DD	Long Rate
base	80	58.8%	1.389	4.40	\$3,522	-\$1,640	65.0%
jepa_only	110	59.1%	1.333	3.86	\$4,245	-\$3,270	52.7%
base_jepa	105	68.6%	1.945	8.70	\$9,132	-\$1,522	63.8%

Table 2: Strict OOS fixed-hold 180 minute proxy backtest.

3.3 Trade Results

The combined model improves every practical trade metric versus the base model. It increases trade count from 80 to 105, win rate from 58.8% to 68.6%, profit factor from 1.389 to 1.945, and PnL from \$3,522 to \$9,132.

4 Per-Ticker OOS Direction Results

Ticker	Trades	WR	PF	Avg bps	PnL	Max DD
QQQ	35	62.9%	1.573	7.76	\$2,717	-\$1,522
SPX	36	72.2%	4.144	14.00	\$5,041	-\$579
SPY	34	70.6%	1.414	4.04	\$1,374	-\$1,026

Table 3: base_jepa strict OOS proxy results by ticker.

SPX is much stronger than SPY and QQQ in PF even though SPX and SPY are highly correlated. The likely explanation is execution and feature microstructure rather than direction alone. SPX option-chain exposure, model thresholds, realized range, and notional scaling produce a different payoff distribution. Correlation of spot returns does not imply identical option-market state or identical threshold behavior.

5 Cost Sensitivity

The edge is execution-sensitive. At 10 bps it disappears. Live deployment should therefore focus on fills, spreads, and avoiding poor-liquidity conditions.

Cost	Trades	WR	PF	Avg bps	PnL
1 bps	105	68.6%	1.945	8.70	\$9,132
3 bps	105	65.7%	1.679	6.70	\$7,032
5 bps	105	64.8%	1.445	4.70	\$4,932
10 bps	105	55.2%	0.976	-0.30	-\$318

Table 4: base_jepa OOS cost sensitivity.

6 0DTE Option Policy OOS Test

6.1 Protocol

The option-policy OOS benchmark uses the frozen-March base_jepa model as the entry generator. For every accepted signal, the simulator constructs option candidates for delta buckets:

$$\Delta \in \{0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70\}.$$

The option side is call for long and put for short. Entry premium is spread-adjusted. Contract count is computed from risk capital. Exits use hard stop and take-profit unless an experiment explicitly evaluates learned exit or oracle exit.

6.2 Aggregate OOS Results

Policy	Trades	WR	PF	PnL	Max DD	Avg PnL	Avg Hold
fixed $\Delta = 0.60$ hard	105	51.4%	1.885	\$25,226	-\$3,354	\$240	152.6
fixed $\Delta = 0.70$ hard	105	56.2%	2.033	\$28,115	-\$3,548	\$268	161.0
learned delta regression hard	105	53.3%	1.889	\$26,223	-\$3,548	\$250	158.6
learned delta ranker hard	105	55.2%	1.965	\$27,145	-\$3,548	\$259	158.5
supervised entry/strike skip hard	46	58.7%	2.235	\$19,472	-\$2,389	\$423	149.9
supervised entry/strike learned exit	46	58.7%	2.489	\$23,381	-\$2,389	\$508	131.7
oracle best delta hard	105	56.2%	5.053	\$69,480	-\$1,703	\$662	161.7
oracle best delta oracle exit	105	87.6%	140.828	\$167,995	-\$342	\$1,600	109.6

Table 5: April–May OOS 0DTE option-policy results.

Fixed $\Delta = 0.70$ is the production choice because it has the strongest combination of volume, PnL, and simplicity among deployable policies in this OOS slice. The supervised skip model has higher PF but only 46 trades, so it is not a direct replacement for the high-volume contract.

6.3 Fixed $\Delta = 0.70$ by Ticker

Ticker	Trades	WR	PF	PnL	Max DD	Avg Delta
QQQ	35	51.4%	1.474	\$3,229	-\$2,144	0.702
SPX	36	63.9%	2.406	\$20,298	-\$3,548	0.703
SPY	34	52.9%	1.770	\$4,589	-\$2,261	0.701

Table 6: April–May OOS fixed-delta option results by ticker.

7 OptionValueJEPa Walk-Forward

7.1 Protocol

OptionValueJEPa is evaluated month by month from August 2023 through May 2026 after at least 12 months of training data. The aggregate experiment includes:

- 2,100 evaluated trades,
- 19,558 candidate option rows,
- 704,088 5 minute dynamic state rows.

7.2 Aggregate Results

Policy	Trades	WR	PF	PnL	Max DD	Avg Delta
fixed $\Delta = 0.70$ hard	2,100	74.3%	5.761	\$1,171,610	-\$5,546	0.703
fixed $\Delta = 0.70$ learned exit 5m	2,100	75.1%	5.398	\$1,055,623	-\$5,546	0.703
option-value hold180 select hard	2,100	65.8%	4.414	\$1,199,159	-\$5,546	0.572
option-value rule select hard	2,100	66.0%	4.415	\$1,187,154	-\$5,546	0.578
option-value best select hard	2,100	51.8%	2.937	\$1,106,884	-\$8,180	0.399
option-value best select learned exit 5m	2,100	52.9%	2.829	\$1,028,827	-\$8,769	0.399
oracle best delta hard	2,100	74.5%	13.993	\$2,084,499	-\$2,093	0.531
oracle best delta oracle exit	2,100	95.9%	643.432	\$4,256,509	-\$342	0.387

Table 7: OptionValueJEPa walk-forward aggregate results.

OptionValueJEPa is not useless: some selector modes produce high PnL. However, the fixed-delta baseline remains stronger in win rate and profit factor. The learned exit variants reduce PnL or PF relative to hard exits. Therefore OptionValueJEPa remains research-only.

8 GBT+JEPa Learned Exit Experiment

Policy	Trades	WR	PF	PnL	Max DD	Avg Hold
fixed 180m	105	68.6%	1.945	\$9,132	-\$2,093	180.0
pooled continuation edge L1	105	68.6%	1.945	\$9,132	-\$2,093	180.0
pooled continuation edge Q75	105	70.5%	1.901	\$8,679	-\$2,093	175.9
per-ticker continuation edge L1	105	68.6%	1.799	\$7,357	-\$2,033	149.7
oracle exit	105	90.5%	44.155	\$27,428	-\$215	110.0

Table 8: Continuation-exit experiment on top of base_jepa.

The oracle exit shows large theoretical headroom. The learned exits failed because they did not reliably estimate continuation value in OOS. Their state features are path-dependent and noisy, and the future-best target has heavy tails.

9 Production Final Diagnostics

9.1 Important Warning

The production final model is trained on all available data through the source parquet maximum date. It is the model to deploy, but its diagnostics are not OOS. They are useful for verifying that the training run completed and the artifacts are internally consistent.

Mode	Trades	WR	PF	Avg bps	PnL	Max DD
base	2,803	87.5%	20.904	30.88	\$865,707	-\$3,111
jepa_only	2,634	86.3%	11.100	28.29	\$745,029	-\$4,711
base_jepa	2,808	88.1%	24.990	31.23	\$876,958	-\$2,674

Table 9: Production-history diagnostics. These are not OOS results.

A quick May 2026 smoke test of the production artifact produced 56 trades, 82.1% win rate, 18.088 PF, and \$13,377 PnL. This confirms that the model loads and trades, but it is not validation because May data was part of the final production training set.

10 Promotion Matrix

Component	Decision	Reason
base_jepa 180m direction	Promote	Beats base GBT in strict April–May OOS on PF, WR, PnL, and drawdown.
jepa_only direction	Do not promote standalone	Contains signal but underperforms the combined model.
Fixed $\Delta = 0.70$ 0DTE execution	Promote	Best simple deployable option policy in the current OOS slice.
OptionValueJEPa strike selector	Research	High PnL in some modes but weaker PF/WR and drawdown than fixed delta.
Learned 5m exit	Reject for production	Oracle headroom exists, but learned exits do not capture it.
Oracle best delta/exit	Diagnostic only	Uses future information and cannot be deployed.

11 Interpretation

The results do not say that the system is solved. They say that the next production release should use base_jepa for 180 minute direction and fixed $\Delta = 0.70$ for 0DTE execution. The evidence is strongest for the direction layer and weaker for learned option optimization. The research gap is now clear: close the gap between fixed delta and oracle selection without sacrificing OOS profit factor.

12 Recommended Next Tests

1. Extend the strict OOS protocol with future months as they become available.
2. Run per-ticker slippage and spread sensitivity for the fixed-delta option policy.
3. Train OptionValueJEPA with ticker-specific calibration or separate heads.
4. Test rank-based losses for option candidate selection, not only regression to clipped return-on-risk.
5. Keep learned exits behind a promotion gate requiring PF, PnL, and drawdown improvement over fixed hold.

13 Conclusion

The current empirical evidence supports a conservative deployment path. JEPA improves the GBT as a latent feature generator. The fixed-delta option layer remains better than learned option selection. Production diagnostics are strong but must not be confused with OOS proof. The next research objective is to make OptionValueJEPA beat the fixed $\Delta = 0.70$ baseline robustly, not merely approach oracle performance in isolated diagnostics.